ENGR 111
Introduction to Engineering
Class #4

Defining the problem

Turning a wish into a number

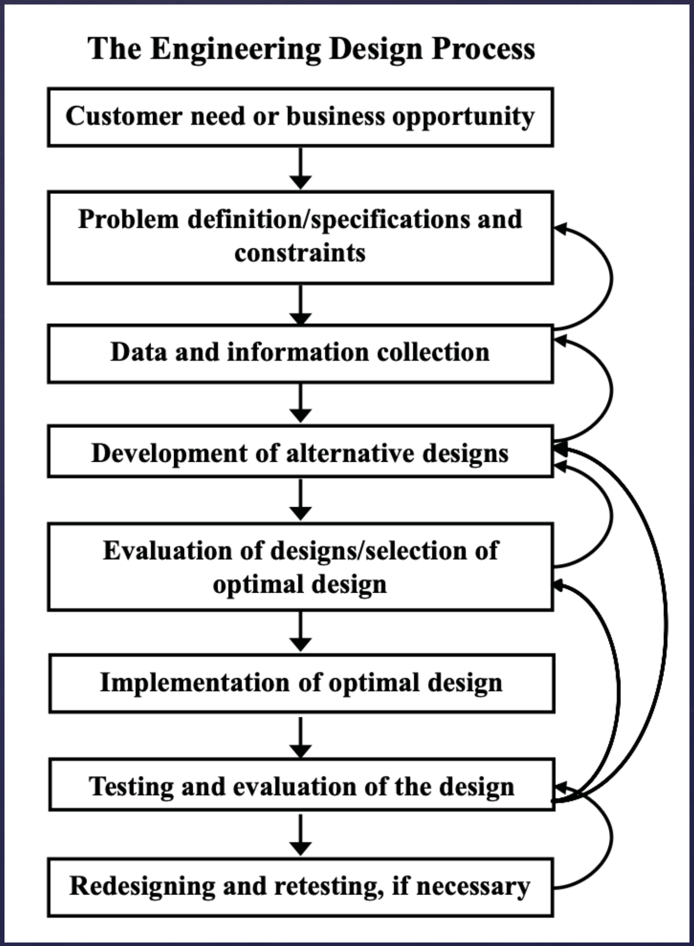
Where we are today

The design process has eight steps. Today we focus on one of them.

Step 2. Problem definition, specifications and constraints.

Everything downstream comes out of it. The alternatives you generate, the criteria you rank them on, the tests you run at the end.

It is the cheapest place to fix anything. Changing a number now costs a conversation. Changing it after you have built the thing costs the build.



Get step 2 wrong and the other seven run perfectly, toward the wrong thing.

"Make it better" or “Make it work” is not a good problem definition

Step 2 is where many projects quietly fail. Nobody notices until step 6.

The request you get will be vague. That is not the customer being lazy. They may not be able to describe the problem precisely, and sometimes they do not know they have it at all.

Your job is to translate it. Turn what they said into something a design can be tested against.

The test of a good problem statement. If two engineers could read yours and build completely different things, you are not finished writing it.

It is a draft, not a contract. Once you find out what is possible you will narrow some numbers and widen others.

What is the vaguest instruction anyone has ever handed you?

A specification has a number and a unit

"Affordable" is a wish. "Forty dollars or less" is a specification.

"Lightweight" is a wish. "Under two kilograms" is a specification.

"Easy to use" is a wish. "A new user completes setup in under three minutes without reading anything" is a specification.

A human-powered helicopter prize. Sixty seconds in the air. Three meters of altitude. Inside a ten-meter square. Those are the specifications. "Fly a helicopter on human power" was only the need.

Choosing the number is the work. It is a design decision, not a clerical one, and it decides much of what follows.

Expect to argue about the numbers. That argument is part of the actual design work.

Specifications are connected

Move one and you move others. This is the part that takes practice.

They have to obey physics. You can write down anything. Only some of it is possible. Lighter, stronger, cheaper and faster all at once usually is not.

They pull on each other. A bigger battery adds weight. More weight needs a stronger frame. A stronger frame costs more and adds weight again. Follow the chain before you commit to a number.

Fewer dependencies is better. If a number can move without dragging five others behind it, the design is far easier to change later.

This is where experience shows. Physics, mathematics, and having built something before. Nobody writes a good set of specifications on their first try.

Ask what else has to change before you write a number down.

An EV charger: one number sets the rest

Choose the power level and most of the rest follows.

Power. 3.3 kW or 100 kW. Sets the magnetics, semiconductors and conductors, and with them size, weight and cost.

AC or DC. A wall in a garage, or a site with a supply big enough to feed it.

Voltage. 400 V and 800 V packs exist. 200 V do not. Inherited from cars you did not design.

Plug, then protocol. CCS: powerline over the pilot wire, ISO 15118 above it. CHAdeMO: CAN.

Cooling. Past a point, passive is not enough. Now you specify a cooling system too.

Protection and compliance. Over-current, short circuit, ground fault, isolation. Then emissions limits, and its own supply has to behave.

Siting. Garage wall or public pedestal: foundation, weather, vandalism, cable reach. Home needs no payment. Public needs a screen and an account.

Which engineering disciplines does this box need?

Three kinds, and where they come from

Specifications generally fall into one of three kinds. Not all of them are yours to choose.

1. Performance. How well. Weight, size, speed, safety, reliability.

2. Economic. How cheap. Teams write performance numbers, feel productive, and forget this one.

3. Scheduling. How soon. Same problem, and it arrives at the worst moment.

Some are handed to you. Standards, codes and law, specific to your industry. Physics sets the rest: no amount of specifying will get you a perpetual motion machine.

Some you will not know about yet. A requirement can stay invisible until you build it, test it, or try to sell it. Interference limits catch first-time hardware teams constantly.

Constraint or criterion?

Both come out of step 2, and they do different jobs.

Constraint

Pass or fail. A design either meets it or it is out of the running.

"Must cost forty dollars or less."

"Must fit through a two-inch pipe."

Many are not yours to set: safety, legal, environmental.

Criterion

How you choose between the designs that passed. Better and worse, not yes and no.

"Cheaper is better."

"Lighter is better."

Useless unless you have more than one design to compare.

Constraints thin the field. Criteria pick the winner out of what is left.

Which is why the process asks you for several designs. With only one, your criteria never do anything.

Who is it for?

We engineer things for people. Name them specifically.

"People" is not an answer (to generic). A chair for a lecture hall and a chair for a dentist are both chairs but share very different specifications based on the application and people it serves.

Go narrow on purpose. Design for one specific, demanding user and the result is often better for everybody. Curb cuts were cut for wheelchairs, and are now used by anyone with a suitcase, a stroller or a delivery trolley. Look up “curb cut effect”.

Somebody is always the beneficiary. Customer, client, user, patient, operator, the public. If nobody is better off, it is not engineering.

Identify the people that will use or benefit from your design (i.e., the users/customers/clients) before you develop the specifications.

Worked example: a hospital water bottle

The customer, specifically. Nurses on a twelve-hour shift, refilling at one shared station, carrying it between rooms.

Three constraints. Holds at least 750 ml. Outer diameter 73 mm or less, so it fits the carts. Survives a 1.5 meter drop onto tile.

Three criteria, ranked. Easy to clean first. Then light. Then cheap. The ranking is a decision, and it is arguable.

What we would go and find out. Measure the cart holders. Ask three nurses what actually breaks. Look up the ward's cleaning standard.

Four questions, in that order. Having a reason for each number matters more than getting it exactly right.